

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Cytarabine Injection BP

CYTARINE

SAFETY WARNINGS:

Only physicians experienced cancer chemotherapy should use Cytarabine Injection.

For induction therapy patients should be treated in a facility with laboratory and supportive resources sufficient to monitor drug tolerance and protect and maintain a patient comprised by drug toxicity. The main toxic effect of cytarabine injection is bone marrow suppression with leukopenia, thrombocytopenia and anemia. Less serious toxicity includes nausea, vomiting, diarrhea and abdominal pain, oral ulceration and hepatic dysfunction.

The physician must judge possible benefit to the patient against known toxic effects of this drug in considering the advisability of therapy with Cytarabine Injection. Before making this judgment or beginning treatment the physician should be familiar with the following text.

DESCRIPTION:

Cytarabine Injection BP, an antineoplastic, is a sterile solution of cytarabine for intravenous, or subcutaneous administration. Cytarabine is a cell cycle-phase specific antimetabolite category of antineoplastics. It is an analogue of the nucleoside cytosine and essentially imitates this natural substance.

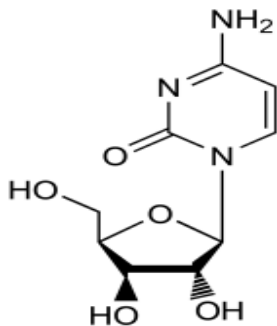
COMPOSITION:

Each ml contains:

Cytarabine BP	100 mg
Water for Injections BP	q.s.

CHEMICAL STRUCTURE:

Cytarabine is chemically 1-β-D-Arabinofuranosylcytosine having molecular formula of $C_9H_{13}N_3O_5$ and molecular weight of 243.22. The chemical structure is:



Cytarabine is white or almost white crystalline powder which is freely soluble in water and very slightly soluble in alcohol and in methylene chloride.

PHARMACOLOGY:

Pharmacodynamics

Mechanism of action:

The exact mechanism(s) of action of cytarabine has not been fully elucidated; however it appears to act through DNA synthesis inhibition. Cytarabine is cytotoxic to a wide variety of proliferating mammalian cells in culture. It exhibits cell phase specificity, primarily killing cells undergoing DNA synthesis (S-phase) and under certain conditions blocking the progression of cells from the G1 phase to the S-phase. A limited, but significant, incorporation of cytarabine into both DNA and RNA has also been reported. Extensive chromosomal damage, including chromatoid breaks, have been produced by cytarabine and malignant transformation of rodent cells in culture has been reported. Deoxycytidine prevents or delays (but does not reverse) the cytotoxic activity.

Cytarabine is converted intracellularly to an active metabolite (cytarabine triphosphate) which inhibits DNA synthesis. The enzyme responsible for this conversion is deoxycytidine kinase which is found predominantly in the liver and possibly the kidney. Cytarabine is inactivated by the enzyme cytidine deaminase found in the intestine, kidney and liver. The ratio of the activating enzyme (deoxycytidine kinase) to the inactivating enzyme (cytidine deaminase) in cells, determines the susceptibility of the tissue to the cytotoxic effects of cytarabine. Tissues with a high susceptibility have high levels of the activating enzyme. Cytarabine has no effect on non-proliferating cells, or on proliferating cells unless in the S or DNA synthesis phase. Thus, cytarabine is a cell cycle phase-specific antineoplastic drug.

Immunosuppressive action

Cytarabine Injection is capable of obliterating immune responses in man during administration with little or no accompanying toxicity. Suppression of antibody responses to *E-coli*-VI antigen and tetanus toxoid have been demonstrated. This suppression was obtained during both primary and secondary antibody responses.

Cytarabine also suppressed the development of cell-mediated immune responses such as delayed hypersensitivity skin reaction to dinitrochlorobenzene. However, it had no effect on already established delayed hypersensitivity reactions.

Following 5-day courses of intensive therapy with cytarabine the immune response was suppressed, as indicated by the following parameters: macrophage ingress into skin windows; circulating antibody response following primary antigenic stimulation; lymphocyte blastogenesis with phytohemagglutinin. A few days after termination of therapy there was a rapid return to normal.

Pharmacokinetics:

Absorption

Orally, less than 20% of a dose of cytarabine is absorbed from the gastrointestinal tract and is ineffective by this route. Subcutaneously or intramuscularly, tritium labelled cytarabine produces peak plasma concentrations of radioactivity within 20 - 60 minutes and are considerably lower than those attained after intravenous administration. Continuous

intravenous infusions produce relatively constant plasma levels in 8 - 24 hours.

Distribution

Cytarabine is widely distributed into tissues including liver, plasma and peripheral granulocytes. Cytarabine crosses the blood brain barrier to a limited extent and is thought to cross the placental barrier. It is not known if cytarabine is distributed into milk.

Cerebrospinal fluid levels of cytarabine are low in comparison to plasma levels after single intravenous injection. However, in one patient in whom cerebrospinal levels were examined after 2 hours of constant intravenous infusion, levels approached 40 percent of the steady state plasma level. With intrathecal administration, levels of cytarabine in the cerebrospinal fluid declined with a first order half-life of about 2 hours. Because cerebrospinal fluid levels of deaminase are low, little conversion to ara-U was observed.

Metabolism

Intravenous doses of cytarabine exhibit a biphasic elimination, with an initial distribution half-life of about 10 minutes during which time a major portion of the drug is metabolised in the liver to the inactive metabolite uracil arabinoside. The secondary elimination half-life is longer, approximately 1 - 3 hours. Metabolism occurs also in the kidneys, gastrointestinal mucosa, granulocytes and other tissues.

Excretion

Cytarabine is mainly excreted via the kidney with 70 - 80% of a dose administered by any route appearing in the urine within 24 hours; approximately 90% as the metabolite and 10% as unchanged drug.

INDICATIONS:

Cytarabine is indicated primarily for:

Induction and maintenance of remission in acute myelocytic leukaemia of both adults and children.

It has also been found to be useful in the treatment of other leukaemias such as:

- Acute lymphocytic leukaemia
- Chronic myelocytic leukaemia (blast phase)

Cytarabine may be used alone or in combination with other antineoplastic agents, the best results are often obtained with combination therapy.

Children with non-Hodgkin's lymphoma have benefited from a combination drug program (LSA2L2) that includes cytarabine.

Remissions induced by cytarabine not followed by maintenance treatment have been brief. Maintenance therapy has extended these and provided useful and comfortable remissions with relatively little toxicity.

ROUTE OF ADMINISTRATION:

Intravenous injection or infusion or subcutaneously.

DOSAGE AND ADMINISTRATION

Cytarabine may be administered by intravenous injection or infusion, or subcutaneously.

Thrombophlebitis has occurred at the site of drug injection or infusion in some patients, and rarely patients have noted pain and inflammation at subcutaneous injection sites. In most instances, however, the drug has been well tolerated.

Patients can tolerate higher total doses when they receive the drug by rapid intravenous injection as compared with slow infusion. This phenomenon is related to the drug's rapid inactivation and brief exposure of susceptible normal and neoplastic cells to significant levels after rapid injection. Normal and neoplastic cells seem to respond in somewhat parallel fashion to these different modes of administration and no clear-cut clinical advantage has been demonstrated for either.

Dose regimens are usually at the discretion of the attending physician. Clinical and haematological responses and tolerance vary between patients and a dose which gives optimal therapeutic effect with minimum toxicity should be used.

Normal Adult Dosage, Single Agent Therapy: Doses of up to 200 mg/m² daily as a continuous intravenous infusion for five days (120 hours) repeated at approximately two weekly intervals have been used. Modification must be made based on results of daily haematological monitoring. After each five-day treatment, drug therapy should be withdrawn to allow for bone marrow recovery.

Dilutions of cytarabine should be made in Glucose 5% or Sodium Chloride 0.9% or Water for Injection, intravenous infusions to the concentrations range between 0.33 - 0.66 mg/mL. In order to reduce any microbiological hazard, it is recommended that dilution should be effected immediately prior to use and infusion commenced as soon as practicable after preparation of the admixture. Admixture should be completed within 48 hours of preparation and any residue discarded. Any storage should be between 30 °C ± 2 °C, protected from light.

Maintenance of Acute Myelocytic Leukaemia in Adults: Maintenance programs are generally modifications of induction programs. Similar schedules of drug therapy to those used for induction are normally employed. Most programs have a greater interval between courses of therapy during remission maintenance.

Induction and Maintenance of Acute Myelocytic Leukaemia (AML) in Children:

Childhood AML has been shown to respond better than adult AML given similar regimes. Where the adult dosage is given in terms of body weight or surface area, the paediatric dosage may be calculated on the same basis, being adjusted on the consideration of such factors as age, body weight or body surface area.

Conditions Requiring Dosage Adjustment

Myelosuppression: The dose of cytarabine should be modified if signs of severe myelosuppression appear, e.g., consideration of discontinuation of the drug if the polymorphonuclear granulocyte count falls below 1×10^9 L or the platelet count falls below

50×10^9 L.

Combination Therapy: Dosage modifications may have to be made when cytarabine is used in combination with other myelosuppressive drugs. Before instituting a programme of combined therapy, the physician should be familiar with the adverse effects, precautions, contraindications and warnings applicable to all the drugs in the programme.

CONTRAINDICATIONS:

Cytarabine Injection is contraindicated in those patients who are hypersensitive to the drug.

ADVERSE EFFECTS:

Haematological: Myelosuppression: Cytarabine is a potent bone marrow suppressant and anaemia, leucopenia, thrombocytopenia, reduced reticulocytes and megaloblastosis can be expected. The severity of these effects is dose and schedule dependent. Cellular changes in the morphology of bone marrow and peripheral smears can be expected.

Following 5-day constant infusions or acute injections of 50 mg/m^2 to 600 mg/m^2 , white cell depression follows a biphasic course. Regardless of initial count, dosage level, or schedule, there is an initial fall starting the first 24 hours with a nadir at days 7-9. This is followed by a brief rise which peaks around the twelfth day. A second and deeper fall reaches nadir at days 15-24. Then there is a rapid rise to above baseline in the next 10 days. Platelet depression is noticeable at 5 days with a peak depression occurring between days 12-15. Thereupon, a rapid rise to above baseline occurs in the next 10 days.

Gastrointestinal: Nausea and vomiting are common and are more severe following rapid intravenous infusion.

Cytarabine (Ara-C) Syndrome: A cytarabine syndrome characterised by fever, myalgia, bone pain, occasionally chest pain, maculopapular rash, conjunctivitis and malaise has been reported. It usually occurs 6 - 12 hours following drug administration. Corticosteroids have been shown to be beneficial in treating or preventing this syndrome. If the symptoms of the syndrome are deemed treatable, corticosteroids should be contemplated as well as continuation of therapy with cytarabine.

Infectious Complications: Viral, bacterial, fungal, parasitic, or saprophytic infections, in any location in the body, may be associated with the use of cytarabine alone or in combination with other immunosuppressive agents following immunosuppressant doses that affect cellular or humoral immunity. These infections may be mild but can be severe and at times fatal.

Tabulated Summaries of Adverse Effects

The reported adverse reactions are listed below by System Organ Class and by frequency. Frequencies are defined as: Very common ($>10\%$), Common ($>1\%$, $\leq 10\%$), Uncommon ($>0.1\%$, $\leq 1\%$), Rare ($>0.01\%$, $\leq 0.1\%$), and Frequency not known (cannot be estimated from available data).

Table 1: Adverse Effects Table

Infections and Infestations	
Very common	Sepsis, pneumonia, infection ^a
Uncommon	Injection site cellulitis
Blood and Lymphatic System Disorders	
Very common	Bone marrow failure, thrombocytopenia, anaemia, anaemia megaloblastic, leukopenia, reticulocyte count decreased
Immune System Disorders	
Uncommon	Anaphylactic reaction ^b , allergic oedema
Metabolism and Nutrition Disorders	
Common	Anorexia
Nervous System Disorders	
Uncommon	Neurotoxicity, neuritis, dizziness, headache
Eye Disorders	
Uncommon	Conjunctivitis ^c
Cardiac Disorders	
Frequency not known	Pericarditis, sinus bradycardia
Vascular Disorders	
Common	Bleeding (all sites), thrombophlebitis
Respiratory, Thoracic and Mediastinal Disorders	
Uncommon	Sore throat, shortness of breath
Gastrointestinal Disorders	
Very common	Stomatitis, mouth ulceration, anal ulcer, anal inflammation, diarrhoea, vomiting ^d , nausea ^d , abdominal pain
Uncommon	Oesophagitis, oesophageal ulceration, bowel necrosis
Frequency not known	Pancreatitis
Hepatobiliary Disorders	
Very common	Hepatic function abnormal
Uncommon	Jaundice
Skin and Subcutaneous Tissue Disorders	
Very common	Alopecia, rash
Common	Skin ulceration
Uncommon	Palmar-plantar erythrodysesthesia syndrome, urticaria, pruritus, freckling
Musculoskeletal, Connective Tissue and Bone Disorders	
Very common	Cytarabine syndrome
Renal and Urinary Disorders	
Uncommon	Renal dysfunction, urinary retention
General Disorders and Administration Site Conditions	
Very common	Pyrexia
Uncommon	Chest pain
Frequency not known	Injection site reaction ^e
Investigations	
Very common	Biopsy bone marrow abnormal, blood smear test abnormal
^a May be mild, but can be severe and at times fatal.	
^b Resulting in cardiopulmonary arrest has been reported following intravenous administration.	
^c May occur with rash and may be hemorrhagic with high dose therapy.	

^dNausea and vomiting are most frequent following rapid intravenous injection.

^ePain and inflammation at subcutaneous injection site.

Table 2: Adverse Effects Table (Experimental Dose Therapy)	
Infections and Infestations	
Frequency not known	Sepsis, liver abscess
Psychiatric Disorders	
Frequency not known	Personality change ^a
Nervous System Disorders	
Very common	Cerebral disorder, cerebellar disorder, somnolence
Frequency not known	Coma, convulsion, peripheral motor neuropathy, peripheral sensory neuropathy
Eye Disorders	
Very common	Haemorrhagic conjunctivitis, corneal disorder
Cardiac Disorders	
Frequency not known	Cardiomyopathy ^b , sinus bradycardia
Respiratory, Thoracic and Mediastinal Disorders	
Very common	Acute respiratory distress syndrome, pulmonary oedema
Gastrointestinal Disorders	
Common	Necrotising colitis
Frequency not known	Bowel necrosis, gastrointestinal ulcer, pneumatosis intestinalis, peritonitis
Hepatobiliary Disorders	
Frequency not known	Liver injury, hyperbilirubinaemia
Skin and Subcutaneous Tissue Disorders	
Common	Skin rash leading to desquamation, alopecia
^a Personality change was reported in association with cerebral and cerebellar dysfunction.	
^b With subsequent death.	

Other Adverse Reactions

Experimental Dose Schedule

A syndrome of sudden respiratory distress, rapidly progressing to pulmonary oedema and a radiographically pronounced cardiomegaly has been reported following experimental high dose therapy with cytarabine used for the treatment of relapsed leukemia; fatal outcome has been reported.

Intermediate Dose Schedule

A diffuse interstitial pneumonitis without clear cause that may have been related to cytarabine was reported in patients treated with experimental intermediate doses of cytarabine (1 g/m²) with and without other chemotherapeutic agents (meta-AMSA, daunorubicin, VP-16).

Interactions with other medicines and other forms of interactions

Methotrexate

Intravenous cytarabine given concomitantly with intrathecal methotrexate may increase the risk of severe neurological adverse reactions such as headache, paralysis, coma and stroke like episodes.

Cytarabine has been reported to inhibit the cellular uptake of methotrexate, thus reducing its effectiveness. Conversely, methotrexate has been reported to reduce the cellular activity of cytarabine. These factors should be taken into consideration if the two drugs are used concomitantly.

Digoxin

Reversible decreases in steady-state plasma digoxin concentrations and renal glycoside excretion were observed in patients receiving beta-acetyldigoxin and chemotherapy regimens containing cyclophosphamide, vincristine and prednisone with or without cytarabine or procarbazine. Steady-state plasma digitoxin concentrations did not appear to change. Therefore, monitoring of plasma digoxin levels may be indicated in patients receiving similar combination chemotherapy regimens. The utilisation of digitoxin for such patients may be considered as an alternative.

Gentamicin

An *in vitro* interaction study between gentamicin and cytarabine showed a cytarabine related antagonism for the susceptibility of *K. pneumoniae* strains. This study suggests that in patients on cytarabine being treated with gentamicin for a *K. pneumoniae* infection, the lack of a prompt therapeutic response may indicate the need for re-evaluation of antibacterial therapy.

Fluorocytosine

Clinical evidence in one patient showed possible inhibition of fluorocytosine efficacy therapy with cytarabine. This may be due to potential competitive inhibition of its uptake.

PRECAUTIONS AND WARNINGS:

Cytarabine Injection should not be administered by the intrathecal route.

Cytarabine should be administered only under constant supervision by physicians experienced in therapy with cytotoxic agents and only when the potential benefits of cytarabine therapy outweigh the possible risks. Patients should be treated in a facility with laboratory and supportive resources sufficient to monitor drug tolerance and protect and maintain a patient compromised by drug toxicity. Appropriate facilities should be available for adequate management of complications should they arise.

The main toxic effect of cytarabine is bone marrow suppression with leukopenia, thrombocytopenia and anaemia. Less serious toxicity includes nausea, vomiting, diarrhoea and abdominal pain, oral ulceration, and hepatic dysfunction.

Myelosuppression

Cytarabine is a potent bone marrow suppressant and the severity depends on the dose of the drug and schedule of administration. Therapy should be started cautiously in patients with pre-existing drug-induced bone marrow suppression. Patients should undergo close medical supervision including daily assessment of leucocyte and platelet levels. Bone marrow examinations should be performed frequently after blasts have disappeared from the peripheral blood. Facilities should be available for management of complications, possibly fatal, of bone marrow suppression (infection resulting from granulocytopenia and other impaired body defences and haemorrhage secondary to thrombocytopenia). Consider suspending or modifying therapy when drug-induced marrow depression has resulted in a platelet count under 50×10^9 L or a polymorphonuclear granulocyte count under 1×10^9 L. Counts of formed elements in the peripheral blood may continue to fall after the drug is stopped and reach lowest values after drug-free intervals of 12 to 24 days. When indicated, restart therapy when definite signs of marrow recovery appear (on successive bone marrow studies). Patients whose drug is withheld until "normal" peripheral blood values are attained may escape from control.

Hepatic and/or Renal Effects

The liver is the main site of inactivation of cytarabine and the normal dosage regimen should be used with caution in patients with pre-existing liver dysfunction or poor renal function. In particular, patients with renal or hepatic function impairment may have a higher likelihood of CNS toxicity after high-dose treatment with cytarabine.

Monitoring

Periodic checks of bone marrow, liver and kidney functions should be performed in patients receiving cytarabine.

Neurological

Cases of severe neurological adverse reactions that ranged from headache to paralysis, coma and stroke-like episodes have been reported mostly in juveniles and adolescents given intravenous cytarabine in combination with intrathecal methotrexate.

Hyperuricaemia

Like other cytotoxic drugs, cytarabine may induce hyperuricaemia secondary to rapid lysis of neoplastic cells. The clinician should monitor the patient's blood uric acid level and be prepared to use such supportive and pharmacological measures as might be necessary to control this problem.

Anaphylaxis

Anaphylactic reactions have occurred with cytarabine treatment. Anaphylaxis that resulted in acute cardiopulmonary arrest and required resuscitation has been reported. This occurred immediately after intravenous administration of cytarabine.

Acute Pancreatitis

Acute pancreatitis has been reported to occur in patients being treated with cytarabine who have had prior treatment with L-asparaginase.

Immunosuppressant Effects/Increased Susceptibility to Infections

Administration of live or live-attenuated vaccines in patients immunocompromised by chemotherapeutic agents including cytarabine, may result in serious or fatal infections. Vaccination with a live vaccine should be avoided in patients receiving cytarabine. Killed or inactivated vaccines may be administered; however, the response to such vaccines may be diminished.

Vomiting

When large intravenous doses are given quickly, patients are frequently nauseated and may vomit for several hours post-injection. The severity is less if the solution is infused.

Conventional Dose Schedules

Abdominal tenderness (peritonitis) and guaiac positive colitis, with concurrent neutropenia and thrombocytopenia, have been reported in patients treated with conventional doses of cytarabine in combination with other drugs. Patients have responded to nonoperative medical management. Delayed progressive ascending paralysis resulting in death has been reported in children with AML following intrathecal and intravenous cytarabine at conventional doses in combination with other drugs.

Experimental Doses

Severe and at times fatal CNS, GI and pulmonary toxicity (different from that seen with conventional therapy regimens of cytarabine) have been reported following some experimental dose schedules of cytarabine. These reactions include reversible corneal toxicity, and haemorrhagic conjunctivitis (which may be prevented or diminished by prophylaxis with a local corticosteroid eye drop); usually reversible cerebral and cerebellar dysfunction (including personality changes, somnolence and coma); severe gastrointestinal ulceration (including pneumatosis cystoides intestinalis leading to peritonitis); sepsis and liver abscess; pulmonary oedema, liver damage with increased hyperbilirubinaemia; bowel necrosis; and necrotising colitis.

Severe sometimes fatal pulmonary toxicity, adult respiratory distress syndrome and pulmonary oedema have occurred following high dose schedules with cytarabine therapy. A syndrome of sudden respiratory distress, rapidly progressing to pulmonary oedema and radiographically pronounced cardiomegaly has been reported following experimental high dose therapy with cytarabine used for the treatment of relapsed leukaemia. The outcome of this syndrome can be fatal.

Cases of cardiomyopathy with subsequent death have been reported following experimental high dose therapy with cytarabine and cyclophosphamide therapy when used for bone marrow transplant preparation. This may be schedule dependent.

Peripheral motor and sensory neuropathies after consolidation with high dose cytarabine, daunorubicin and asparaginase have occurred in adult patients with non-lymphocytic leukaemia. Patients treated with high dose cytarabine should be observed for neuropathy since dose schedule alterations may be needed to avoid irreversible neurologic disorders.

Rarely, severe skin rash, leading to desquamation has been reported. Complete alopecia is more commonly seen with experimental high dose therapy than with standard cytarabine

treatment programs.

Use in the elderly

No data available

Paediatric use

No data available

FERTILITY, PREGNANCY AND LACTATION:**Effects on fertility**

No data available

Use in pregnancy

Cytarabine is known to be teratogenic in some animal species and its use in pregnant women is not recommended. Cytarabine should only be used in women of child-bearing potential if the expected benefits outweigh the risks of therapy and adequate contraception is used.

A review of the literature has shown 32 reported cases where cytarabine was given during pregnancy, either alone or in combination with other cytotoxic agents: Eighteen normal infants were delivered. Four of these had first trimester exposure. Five infants were premature or of low birth weight. Twelve of the 18 normal infants were followed up at ages ranging from 6 weeks to 7 years, and showed no abnormalities. One apparently normal infant died at 90 days of gastroenteritis.

Two cases of congenital abnormalities have been reported, one with upper and lower distal limb defects, and the other with extremity and ear deformities. Both of these cases had first trimester exposure. There were seven infants with various problems in the neonatal period, including pancytopenia; transient depression of WBC, haematocrit or platelets; electrolyte abnormalities; transient eosinophilia; and one case of increased IgM levels and hyperpyrexia possibly due to sepsis. Six of the seven infants were also premature. The child with pancytopenia died at 21 days of sepsis.

Therapeutic abortions were done in five cases. Four fetuses were grossly normal, but one had an enlarged spleen, and another showed Trisomy C chromosome abnormality in the chorionic tissue.

Because of the potential for abnormalities with cytotoxic therapy, particularly during the first trimester, a patient who is or who may become pregnant while on cytarabine should be apprised of the potential risk to the foetus and the advisability of pregnancy continuation. There is a definite, but considerably reduced risk if therapy is initiated during the second or third trimester. Although normal infants have been delivered to patients treated in all three trimesters of pregnancy, follow-up of such infants would be advisable.

Use in lactation

It is not known whether cytarabine is excreted in breast milk so breast feeding should be discontinued during cytarabine therapy in lactating women.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

The effects of this medicine on a person's ability to drive and use machines were not assessed.

OVERDOSE:

There is no antidote for overdosage of cytarabine. Doses of 4.5 g/m² by intravenous infusion over 1 hour every 12 doses for 12 doses has caused an unacceptable increase in irreversible CNS toxicity and death. Symptoms of overdose include nausea, vomiting, diarrhoea, ulceration and bleeding of the gastrointestinal tract, myelosuppression, severe skin rash, CNS toxicity (including cerebral and cerebellar dysfunction), cardiac disorders, pulmonary and corneal toxicity, fever, myalgia, bone pain, chest pain and conjunctivitis.

STORAGE:

Store below 25°C. Protect from light.

SHELF-LIFE:

18 months

INCOMPATIBILITIES:

Cytarabine must not be mixed with other medicinal products except those mentioned in Dose and method of administration. Cytarabine has been known to be physically incompatible with heparin, insulin, fluorouracil, penicillins such as oxacillin and penicillin G sodium, and methylprednisolone sodium succinate.

HANDLING AND DISPOSAL:

Procedures for proper handling and disposal of anti-cancer drugs should be considered. Several guidelines on this subject have been published. There is no general agreement that all of the procedures recommended in the guidelines are necessary or appropriate.

Parenteral drugs should be inspected visually for particulate matter and discoloration, prior to administration, whenever solution and container permit.

Special Precautions for Handling

If crystallization is observed as a result of exposure to low temperatures, redissolve the crystals by warming up to 55°C for no longer than 30 minutes and shake until the crystals are dissolved. Allow to cool prior to use.

PRESENTATION:

Cytarabine Injection BP is available as 1 ml in 2 ml, 5 ml and 10 ml vial containing 100 mg, 500 mg and 1 g of Cytarabine BP, respectively.

REFERENCES:

Product Information, Cytarabine CS (Cytarabine, 100mg) Solution for Injection (Cytarabine-0521); Pfizer (Perth) Pty Limited, 24 May 2021.

MANUFACTURER INFORMATION:

Manufactured in India by:

Fresenius Kabi Oncology Limited

Village - Kishanpura,

P.O. - Guru Majra

Tehsil - Nalagarh,

Distt. - Solan (H.P.)-174101

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